Ahmed Mohamed Ragab

PhD Candidate in Institute of Bioinformatics and Structural Biology, National Tsing Hua University, Taiwan
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EDUCATION

PhD in Bioinformatics and Structural Biology **Feb. 2022 – Present**
Institute of Bioinformatics and Structural Biology, National Tsing Hua University (NTHU), Hsinchu, Taiwan
Supervisors: Prof. Lee-Wei Yang

MSc in Bioinformatics and Structural Biology **Sep. 2019 – Jan. 2022**
Institute of Bioinformatics and Structural Biology, National Tsing Hua University (NTHU), Hsinchu, Taiwan
Supervisors: Prof. Lee-Wei Yang and Prof. Jin-Der Wen
Thesis: Accurate and Efficient Chemical Similarity Search Tool and Contact-Distribution-Matching Method Jointly Identify FDA-Approved Drugs That Modulate SARS-CoV-2 -1 PRF and Suppress Its Replication

BSc in Zoology/Chemistry **Sep. 2011 - May 2015**
Cairo University, Giza, Egypt

RESEARCH EXPERIENCE

Predoctoral researcher, Computational Biology Laboratory, NTHU, Taiwan **Feb. 2022 – Still**
Project 1: Using Steered Molecular Dynamics to study -1 programmed ribosomal frameshifting mechanism of SARS-CoV2.
Project 2: Acid-Enriched Unfolding Intermediates of a pH-Sensitive Pseudoknot Enhances Ribosomal Frameshifting.
Project 3: Reassessment of Molecular Mechanisms Underlying Transcription Initiation.

Research Assistant, Computational Biology Laboratory, NTHU, Taiwan **Sep. 2019 – Jan. 2022**
Project: Developing a computational method for rational drugging of -1 programmed ribosomal frameshifting of SARS-CoV2.

Research Assistant, X-ray Determination of the Structure of Matter laboratory, Zewail City, Giza, Egypt **June – Aug. 2019**

Research Assistant, Cancer Biology Research Lab, Cairo University, Giza, Egypt **Sep. – Dec. 2016**
Project: Silencing of Syndecan-1 in inflammatory breast cancer cell lines as a novel therapeutic target.

PROFESSIONAL EXPERIENCE

Lab technician, Genome Once Centre, Giza, Egypt **May 2015 – Jul. 2018**
Skills and Techniques: Cytogenetics FISH, Flow Cytometry for Immunophenotyping, DNA/RNA extraction, gel electrophoresis, conventional PCR, RT-qPCR.

PUBLICATIONS

Ragab, A. M., Ortiz, C. L. D., Chang, K.-C., Salawu, E. O., Yang, C.-H., Hsu, D. S.-T., Chang, K.-Y., Yang, L.-W., Wen, J.-D. Acid-Enriched Unfolding Intermediates of a pH-Sensitive Pseudoknot Enhances Ribosomal Frameshifting Efficiency. (**In Preparation**)

Ragab, A. M., Ortiz, C. L. D., Huang, Y.-T., Wang, J.-Z., Wu, Y.-H., Wen, J.-D., Yang, L.-W. (2026). Drug Functional Site-Unknown Molecular Targets Exemplified by SARS-CoV-2 pseudoknot and ribosomal frameshifting-modulators.

bioRxiv, 2026.07.02.735960. https://doi.org/10.64898/2026.07.02.735960

Sanad, A. M. R. M., Chang, S.-Y., Wen, J.-D., & Yang, L.-W. (2026). Drug repurposing and mechanistic dissection of −1 programmed ribosomal frameshifting in SARS-CoV-2. *Biophysical Journal, 125*(4), 256a. https://doi.org/10.1016/j.bpj.2025.11.1646

PRESENTATIONS AND CONFERENCES

Summer RosettaCon 2026 – University of Washington, Seattle, USA (Poster presentation) **Aug. 03 – 07, 2026**
Biophysical Society Annual Meeting 2026 – San Francisco, USA (15-minutes Platform talk and Poster presentation) **Feb. 21 – 25, 2026**
NTHU-Osaka U. Joint Symposium 2025 – Osaka, Japan (Oral presentation) **Sep. 28 – Oct. 01, 2025**
Complex Systems Symposium 2025 – NYCU, Taiwan (Oral presentation) **May 16 – 17, 2025**
Biologic Summit: Demystifying AI/ML for Biologic Drug Development – San Diego, USA (Virtual, Poster presentation) **Jan. 14 – 16, 2025**
8th Cross-Strait Symp. on Biomedical Sciences – CUHK, Hong Kong (Poster presentation) **Nov. 08 – 11, 2023**
6th International Conference on Molecular Simulation – NTU, Taiwan (Poster presentation) **Oct. 06 – 08, 2023**
Nucleic Acids: Prospect and Therapeutic Applications – NTU, Taiwan (Poster presentation) **Jul. 07, 2023**
Department Retreat – Institute of Bioinformatics and Structural Biology, NTHU, Hsinchu, Taiwan (Oral presentation) **Aug. 30 – 31, 2023**
2022 TSBMB Autumn Camp – Hsinchu, Taiwan (Poster presentation) **Nov. 04 – 06, 2022**

COMPUTER AND PROGRAMNING SKILLS

- Programming languages: Python, Bash, R, and Tcl/Tk.
- Statistical analysis: Familiar with R
- Machine learning/Artificial Intelligence: NumPy, Pandas, PyTorch, SciKit-Learn, TensorFlow, Keras.
- Molecular Modeling and Docking: MD simulations, Autodock Vina, Amber, VMD, Chimera, and PyMOL.

SCHOLARSHIPS AND AWARDS

NTHU-Emmanuel O. Salawu Scholarship for PhD Students **2026**
Dr. Emmanuel O. Salawu Travel Grant (Self-instigated, 1st awardee) **2026**
NTHU Office of Academic Affairs Travel Grant **2026**
NTHU Office of Global Affairs International Conference Travel Grant (2nd time) **2026**
National Center for Theoretical Sciences Travel Grant **2026**
NTHU Office of Global Affairs International Conference Travel Grant (1st time) **2026**
National Science and Technology Council (Taiwan) Travel Grant (2nd time) **2025**
Biophysical Society Travel Grant **2025**
NTHU College of Life Sciences and Medicine Travel Grant **2025**
AWS Scholarship for Machine Learning Fundamentals at Udacity **2025**
National Science and Technology Council (Taiwan) Travel Grant (1st time) **2024**
AWS Scholarship for AI Programming with Python at Udacity **2024**
2nd best Poster Presentation Award, 8th Cross-Strait Symposium on Biomedical Sciences **2023**
NTHU College of Life Sciences and Medicine Travel Grant **2023**
1st Best Presentation Award, Department Retreat, NTHU, Taiwan **2023**
National Tsing Hua University International Students Scholarship for PhD **2022**
Taiwan Government Scholarship of Ministry of Education for master's degree **2019**
National Tsing Hua University Scholarship for master's degree **2019**
Faculty of Science top-student award (Cairo University, Egypt) **2015**